

Accenture - 100 Technical MCQ Questions

Accenture

For BTech Computer Science / IT Students

Object-Oriented Programming

1. In Python, which method is automatically called when an object is created?

- A) `__new__` then `__init__`
- B) `__del__`
- C) `__call__`
- D) `__str__`

2. Which of the following correctly describes 'late binding'?

- A) Method call resolved at compile time
- B) Method call resolved at runtime based on object type
- C) A binding used only for static variables
- D) A binding used for interfaces only

3. Method overloading is an example of which type of polymorphism?

- A) Runtime polymorphism
- B) Compile-time polymorphism
- C) Dynamic binding
- D) Late binding

4. Which of the following is true regarding static methods and polymorphism?

- A) Static methods support runtime polymorphism
- B) Static methods are resolved at compile time and do not support runtime polymorphism
- C) Static methods cannot be called
- D) Static methods must be abstract

5. Which of these is a valid example of encapsulation in practice?

- A) Making instance variables private and exposing getters/setters
- B) Making all variables public
- C) Using global variables everywhere
- D) Avoiding classes altogether

6. What is a 'friend' function in C++?

- A) A function that has access to private/protected members of a class
- B) A public function only
- C) A function that cannot access class members
- D) A default constructor

7. Which keyword is used in Java to prevent a class from being inherited?

- A) `static`
- B) `final`
- C) `private`
- D) `abstract`

8. Which of the following is NOT a pillar of Object-Oriented Programming?

- A) Encapsulation
- B) Polymorphism
- C) Compilation
- D) Inheritance

9. What is operator overloading?

- A) Giving additional meaning to an operator for user-defined types
- B) Removing operators from a language
- C) A compilation error
- D) A runtime exception

10. What is object composition?

- A) Building complex objects by combining simpler objects
- B) Inheriting from multiple classes
- C) Creating static objects
- D) Deleting objects

11. What is the diamond problem in OOP?

- A) An ambiguity arising from multiple inheritance of the same base class
- B) A memory leak issue
- C) A sorting algorithm's worst case
- D) A type of exception

12. What is the output-behavior difference between shallow copy and deep copy of an object?

- A) Shallow copy duplicates only references to nested objects; deep copy duplicates nested objects too
- B) There is no difference
- C) Deep copy is always faster
- D) Shallow copy always throws an exception

13. Which of these is true about interfaces in Java 8+?

- A) They can have default and static methods with bodies
- B) They can have instance variables with values changed at runtime
- C) They support constructors
- D) They cannot be implemented by multiple classes

14. What is encapsulation?

- A) Hiding implementation details and exposing only functionality
- B) Creating multiple classes
- C) Compiling code faster
- D) Using loops effectively

15. Which OOP feature allows a subclass to provide a specific implementation of a method already defined in its superclass?

- A) Overloading
- B) Overriding
- C) Encapsulation
- D) Abstraction

16. Which design principle states 'favor composition over inheritance'?

- A) SOLID-related OOP design guideline
- B) A rule that disallows interfaces
- C) A Java-only rule
- D) A database design rule

DBMS & SQL

17. What is a trigger in SQL?

- A) A special stored procedure automatically executed in response to certain events on a table
- B) A type of index
- C) A join operation
- D) A constraint on primary keys

18. What does DBMS stand for?

- A) Database Management System
- B) Data Backup Management Service
- C) Database Model Structure
- D) Data Base Modeling Software

19. Which SQL function is used to combine strings from multiple rows into one string (varies by RDBMS)?

- A) CONCAT() only
- B) GROUP_CONCAT()/STRING_AGG() depending on RDBMS
- C) MERGE()
- D) JOIN()

20. Which of the following commands removes all rows from a table but keeps its structure, and cannot be rolled back in most databases?

- A) DELETE
- B) DROP
- C) TRUNCATE
- D) REMOVE

21. Which isolation level allows 'dirty reads'?

- A) Serializable
- B) Repeatable Read
- C) Read Committed
- D) Read Uncommitted

22. Which of the following is an example of a NoSQL database?

- A) MySQL
- B) PostgreSQL
- C) MongoDB
- D) Oracle

23. In SQL, what does the LIKE operator do?

- A) Performs pattern matching using wildcards such as % and _
- B) Compares two numeric values
- C) Joins two tables
- D) Groups rows

24. Which of these correctly describes a many-to-many relationship implementation?

- A) Implemented directly with a foreign key in one table
- B) Implemented using a junction/associative table with foreign keys to both tables
- C) Not possible in relational databases
- D) Requires a trigger

25. What is a primary key?

- A) A column that can contain duplicate values
- B) A column (or set of columns) that uniquely identifies each row and cannot be NULL
- C) An index used only for sorting
- D) A key used only in foreign tables

26. Which SQL keyword is used to remove duplicate rows from a result set?

- A) UNIQUE
- B) DISTINCT
- C) FILTER
- D) ONLY

27. Which SQL clause is used to filter groups after a GROUP BY?

- A) WHERE
- B) HAVING
- C) FILTER
- D) ORDER BY

28. What is the main advantage of using an ORM (Object-Relational Mapping) tool?

- A) It eliminates the need for a database entirely
- B) It lets developers interact with the database using objects instead of raw SQL
- C) It automatically normalizes data
- D) It replaces indexes

29. What is a correlated subquery?

- A) A subquery that references a column from the outer query
- B) A subquery that never executes
- C) A subquery used only in INSERT statements
- D) A subquery without a SELECT clause

30. Which SQL command grants specific privileges to a user?

- A) REVOKE
- B) GRANT
- C) ALLOW
- D) PERMIT

31. What is the purpose of database normalization?

- A) To reduce data redundancy and improve data integrity
- B) To make queries slower
- C) To increase storage duplication
- D) To remove all keys

32. What is the purpose of the GROUP BY clause?

- A) To sort data in ascending order
- B) To arrange identical data into groups for aggregate calculations
- C) To filter rows before aggregation
- D) To join multiple tables

33. What is a materialized view?

- A) A view whose result set is physically stored and periodically refreshed
- B) A temporary table only
- C) A synonym for an index
- D) A type of trigger

34. What does 'sharding' mean in database scaling?

- A) Splitting a large database horizontally across multiple servers
- B) Encrypting data
- C) Creating a backup
- D) Normalizing tables

35. What is a schema in a database context?

- A) The logical structure that defines tables, columns, and relationships
- B) A backup file
- C) A stored procedure
- D) A query optimizer

36. Which of the following best explains 'eventual consistency' in distributed/NoSQL databases?

- A) All nodes are always immediately consistent
- B) Given enough time without new updates, all nodes will converge to the same value
- C) Data is never consistent
- D) It refers only to backup consistency

37. Which normal form removes partial dependency on a composite primary key?

- A) 1NF
- B) 2NF
- C) 3NF
- D) BCNF

38. A foreign key is used to:

- A) Uniquely identify a row in its own table
- B) Establish a link between two tables
- C) Encrypt sensitive data
- D) Speed up all queries automatically

Operating Systems

39. Which of the following is an example of external fragmentation?

- A) Free memory scattered in small blocks between allocated partitions, none large enough for a new request
- B) Wasted space within an allocated fixed-size block
- C) A page fault
- D) A deadlock

40. Which scheduling algorithm can cause starvation?

- A) Round Robin
- B) Priority Scheduling (without aging)
- C) FCFS
- D) None of these

41. What does the term 'fragmentation' refer to in memory management?

- A) Wasted memory space due to allocation and deallocation patterns (internal or external)
- B) A type of process priority
- C) A CPU scheduling delay
- D) A file system backup issue

42. What is an inode in a Unix-like file system?

- A) A data structure storing metadata about a file (permissions, size, pointers to data blocks) but not its name
- B) The file's name only
- C) A type of directory
- D) A kernel module

43. What is the role of the dispatcher in CPU scheduling?

- A) To select which process runs next based purely on priority
- B) To give control of the CPU to the process selected by the short-term scheduler
- C) To manage disk I/O only
- D) To handle network requests

44. What is the difference between a hard link and a symbolic link in a file system?

- A) A hard link points directly to the inode; a symbolic link is a separate file containing a path to the target
- B) They are identical
- C) Symbolic links cannot be broken
- D) Hard links can span across different file systems while symbolic links cannot

45. What is a deadlock?

- A) A state where two or more processes are waiting indefinitely for resources held by each other
- B) A process that has finished execution
- C) A type of memory leak
- D) A CPU scheduling technique

46. What is the banker's algorithm used for?

- A) Deadlock avoidance by checking safe states before allocating resources
- B) Memory compression
- C) File compression
- D) CPU scheduling

47. Which of the following best defines 'mutual exclusion' as a deadlock condition?

- A) A resource can be used by multiple processes simultaneously
- B) At least one resource must be held in a non-shareable mode
- C) Processes never request resources
- D) Resources are always preemptable

48. What is the difference between a process and a thread?

- A) A process is a lightweight version of a thread
- B) A thread is an independent execution unit within a process, sharing the process's memory space
- C) Threads have separate memory spaces from their process
- D) There is no difference

49. Which memory management technique divides memory into fixed-size blocks?

- A) Segmentation
- B) Paging
- C) Swapping
- D) Caching

50. Which of the following is NOT one of the four necessary conditions for deadlock?

- A) Mutual Exclusion
- B) Hold and Wait
- C) Preemption
- D) Circular Wait

51. Which file system attribute determines the smallest unit of disk space allocation?

- A) Sector/Block size
- B) File descriptor
- C) Inode number
- D) Directory depth

52. What is the primary function of an operating system?

- A) To compile programs
- B) To manage hardware resources and provide services to applications
- C) To design websites
- D) To encrypt files only

Computer Networks

53. Which layer of the OSI model is responsible for routing?

- A) Data Link Layer
- B) Network Layer
- C) Transport Layer
- D) Session Layer

54. What does a 'ping' command test?

- A) Reachability and round-trip time to a host using ICMP echo requests
- B) DNS resolution only
- C) Bandwidth of a connection
- D) Server uptime history

55. What is the primary purpose of a load balancer?

- A) To distribute incoming network traffic across multiple servers to improve reliability and performance
- B) To encrypt data at rest
- C) To act as a firewall only
- D) To assign static IP addresses

56. How many layers are there in the OSI model?

- A) 5
- B) 6
- C) 7
- D) 8

57. What is the default port number for HTTPS?

- A) 80
- B) 21
- C) 443
- D) 22

58. What is a MAC address?

- A) A unique hardware identifier assigned to a network interface
- B) An IP address used only for routing
- C) A type of DNS record
- D) A session identifier

59. Which protocol operates at the Transport layer and is connection-oriented?

- A) UDP
- B) IP
- C) TCP
- D) ICMP

60. Which of the following best describes a CDN (Content Delivery Network)?

- A) A distributed network of servers that delivers content to users from a location geographically closer to them
- B) A type of firewall
- C) A DNS resolution protocol
- D) A database replication technique

61. Which protocol is used to resolve a domain name to an IP address?

- A) FTP
- B) DNS
- C) SMTP
- D) ARP

62. Which of the following is an application-layer protocol?

- A) TCP
- B) IP
- C) HTTP
- D) Ethernet

63. What is the default port number for HTTP?

- A) 21
- B) 25
- C) 80
- D) 443

64. What is the main difference between TCP and UDP?

- A) TCP is connectionless, UDP is connection-oriented
- B) TCP is connection-oriented and reliable, UDP is connectionless and faster but unreliable
- C) There is no difference
- D) UDP guarantees ordered delivery while TCP does not

Data Structures & Algorithms

65. What is the time complexity of building a heap from an unsorted array (heapify)?

- A) $O(n \log n)$
- B) $O(n)$
- C) $O(\log n)$
- D) $O(n^2)$

66. What is the time complexity of binary search on a sorted array of n elements?

- A) $O(n)$
- B) $O(\log n)$
- C) $O(n \log n)$
- D) $O(1)$

67. Which data structure underlies the implementation of recursion internally?

- A) Queue
- B) Stack (call stack)
- C) Heap (memory heap)
- D) Graph

68. Which of the following is true about a doubly linked list compared to a singly linked list?

- A) It uses less memory per node
- B) Each node has pointers to both the next and previous nodes, allowing traversal in both directions
- C) It cannot be traversed backward
- D) It has no advantages

69. What is the primary difference between a greedy algorithm and dynamic programming?

- A) Greedy makes a locally optimal choice at each step without reconsidering, DP considers overlapping subproblems and often guarantees a globally optimal solution
- B) They are the same technique
- C) DP never uses recursion
- D) Greedy always gives an optimal solution while DP does not

70. Which graph algorithm is used to find the shortest path in a graph with possible negative edge weights (no negative cycles)?

- A) Dijkstra's Algorithm
- B) Bellman-Ford Algorithm
- C) Prim's Algorithm
- D) Kruskal's Algorithm

71. What is memoization?

- A) Storing the results of expensive function calls and reusing them when the same inputs occur again
- B) A sorting technique
- C) A type of recursion without a base case
- D) A method to delete duplicate array elements

72. Which data structure uses FIFO (First In First Out) order?

- A) Stack
- B) Queue
- C) Tree
- D) Graph

73. What is the amortized time complexity of appending an element to a dynamic array (like Python's list or Java's ArrayList)?

- A) $O(n)$
- B) $O(\log n)$
- C) $O(1)$ amortized
- D) $O(n^2)$

74. What is tail recursion?

- A) A recursive call that is the last operation in a function, which can be optimized by some compilers into iteration
- B) A recursive function with no base case
- C) A function that calls itself twice
- D) A type of infinite loop

75. What is the time complexity of the Two-Pointer technique on a sorted array to find a pair with a given sum?

- A) $O(n^2)$
- B) $O(n \log n)$
- C) $O(n)$
- D) $O(\log n)$

76. What is a monotonic stack typically used for?

- A) Finding the next greater/smaller element efficiently
- B) Sorting an array in place
- C) Implementing recursion
- D) Hashing

77. What is the time complexity of Floyd-Warshall algorithm for all-pairs shortest paths?

- A) $O(V^2)$
- B) $O(V^3)$
- C) $O(E \log V)$
- D) $O(V + E)$

78. Which of the following best describes a trie data structure?

- A) A tree-like structure used for efficient retrieval of strings, especially for prefix matching
- B) A type of balanced binary search tree
- C) A hashing algorithm
- D) A graph traversal method

79. Which technique explores all possible options and backtracks when a solution path fails, often used in puzzles like N-Queens?

- A) Greedy algorithm
- B) Backtracking
- C) Dynamic programming
- D) Binary search

80. What is the time complexity of finding the minimum element in an unsorted array?

- A) $O(1)$
- B) $O(\log n)$
- C) $O(n)$
- D) $O(n \log n)$

81. What is the worst-case time complexity of QuickSort?

- A) $O(n \log n)$
- B) $O(n^2)$
- C) $O(\log n)$
- D) $O(n)$

82. Which data structure uses LIFO (Last In First Out) order?

- A) Queue
- B) Stack
- C) Linked List
- D) Array

83. Which algorithm is used to find the Minimum Spanning Tree of a graph?

- A) Dijkstra's Algorithm
- B) Kruskal's or Prim's Algorithm
- C) Bellman-Ford Algorithm
- D) Floyd-Warshall Algorithm

84. Which of these best describes the 'sliding window' technique?

- A) Maintaining a subset of elements (a window) over a sequence and adjusting its bounds to solve subarray/substring problems efficiently
- B) A sorting technique for linked lists
- C) A hashing technique
- D) A tree traversal method

Programming Language Fundamentals

85. What does 'pass by reference' mean?

- A) A copy of the variable's value is passed to the function
- B) The function receives a reference to the actual variable, so changes affect the original
- C) Variables cannot be passed to functions
- D) Only primitive types can be passed

86. In Python, which of the following is a mutable data type?

- A) Tuple
- B) String
- C) List
- D) int

87. What is a lambda function/expression?

- A) A named function stored in a file
- B) An anonymous, inline function typically used for short operations
- C) A type of class
- D) A compiler directive

88. In C++, what does the 'const' keyword do when applied to a variable?

- A) Makes the variable a pointer
- B) Prevents the variable's value from being modified after initialization
- C) Declares a static variable
- D) Declares a global variable

89. What is the purpose of the 'finally' block in exception handling (Java/Python)?

- A) It executes only if an exception occurs
- B) It executes only if no exception occurs
- C) It always executes regardless of whether an exception occurred or not
- D) It is used to declare exceptions

90. What is method chaining?

- A) Calling multiple methods on the same object in a single statement by having each method return the object itself
- B) A way to declare multiple classes
- C) A type of exception handling
- D) A memory management technique

91. Which keyword is used to handle exceptions in Python?

- A) catch
- B) except
- C) rescue
- D) handle

92. In Java, what is the difference between '==' and '.equals()' when comparing objects?

- A) They always behave identically
- B) '==' compares references while '.equals()' compares content (if overridden)
- C) '.equals()' is only for primitives
- D) '==' is only used for Strings

93. Which of the following best describes garbage collection?

- A) A manual process where the developer frees memory explicitly
- B) An automatic process that reclaims memory occupied by objects no longer in use
- C) A compilation step
- D) A type of exception handling

94. What is the difference between a compiler and an interpreter?

- A) A compiler translates the whole program to machine code before execution, an interpreter executes code line-by-line
- B) They are identical
- C) Interpreters are always faster than compilers
- D) Compilers cannot detect syntax errors

Web & Cloud Fundamentals

95. What is the difference between IaaS, PaaS, and SaaS?

- A) They are all the same cloud service model
- B) IaaS provides raw infrastructure, PaaS provides a platform to build apps, SaaS delivers ready-to-use software
- C) SaaS provides infrastructure while IaaS provides software
- D) PaaS is only for storage

96. What does 'statelessness' mean in the context of REST APIs?

- A) The server stores client session state between requests
- B) Each request from a client contains all the information needed to process it, with no reliance on stored server-side session state
- C) The API has no state at all, including no data
- D) Stateless APIs cannot use authentication

97. What is the purpose of cookies in web applications?

- A) To store small pieces of data on the client side to maintain state/session information
- B) To encrypt HTTPS traffic
- C) To speed up DNS resolution
- D) To compress images

98. What does CSS stand for?

- A) Cascading Style Sheets
- B) Computer Style Sheets
- C) Creative Style System
- D) Cascading Software Sheets

99. What is auto-scaling in cloud computing?

- A) Automatically adjusting compute resources up or down based on demand
- B) Manually adding servers only
- C) A type of database indexing
- D) A billing model only

100. What is the purpose of a Content Delivery Network (CDN)?

- A) To cache and serve content from servers geographically closer to users, reducing latency
- B) To encrypt all web traffic
- C) To replace DNS servers
- D) To manage database transactions

Answer Key

1-A	2-B	3-B	4-B	5-A	6-A	7-B	8-C	9-A	10-A
11-A	12-A	13-A	14-A	15-B	16-A	17-A	18-A	19-B	20-C
21-D	22-C	23-A	24-B	25-B	26-B	27-B	28-B	29-A	30-B
31-A	32-B	33-A	34-A	35-A	36-B	37-B	38-B	39-A	40-B
41-A	42-A	43-B	44-A	45-A	46-A	47-B	48-B	49-B	50-C
51-A	52-B	53-B	54-A	55-A	56-C	57-C	58-A	59-C	60-A
61-B	62-C	63-C	64-B	65-B	66-B	67-B	68-B	69-A	70-B
71-A	72-B	73-C	74-A	75-C	76-A	77-B	78-A	79-B	80-C
81-B	82-B	83-B	84-A	85-B	86-C	87-B	88-B	89-C	90-A
91-B	92-B	93-B	94-A	95-B	96-B	97-A	98-A	99-A	100-A

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